



JAIPUR NATIONAL UNIVERSITY

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SCHOOL OF ENGINEERING AND TECHNOLOGY

Mid-Term Test - I, March, 2022

B. Tech. - II Semester

ENGINEERING MECHANICS

BTEBE205

Time: 2 Hours

Max. Marks: 20

Q.1 Attempt any 8 questions:

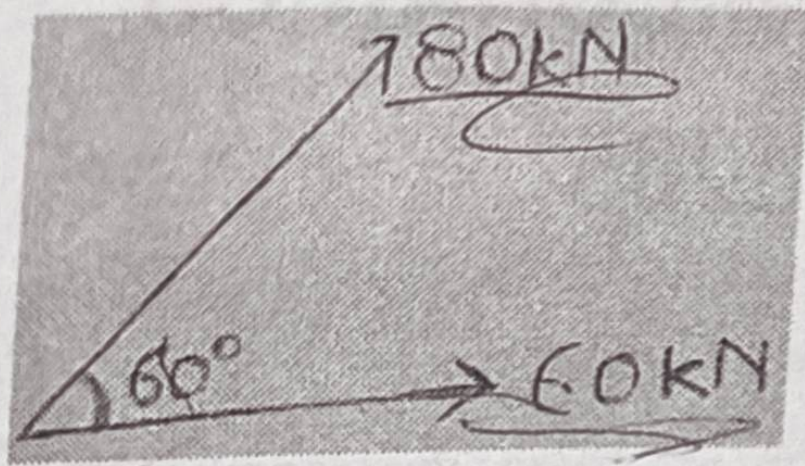
(1×8=8)

- (a) Explain the fundamental Law of Mechanics.
- (b) Write short notes on Lami's Theorem with diagram.
- (c) Write the condition of equilibrium of a body.
- (d) What is Moment of a Force?
- (e) What is Couples of a Force?
- (f) What is the difference between Coplanar and Non-Coplanar forces?
- (g) Mention any 3 points to define the importance of mechanics.
- (h) What do you understand by three force principle?
- (i) What do you understand by principle of transmissibility of force?
- (j) What are the branches of applied mechanics?
- (k) Make a classification chart to define the system of forces.
- (l) Define the principle of virtual work.

Q.2 (a) State and Prove Parallelogram law of forces with diagram.

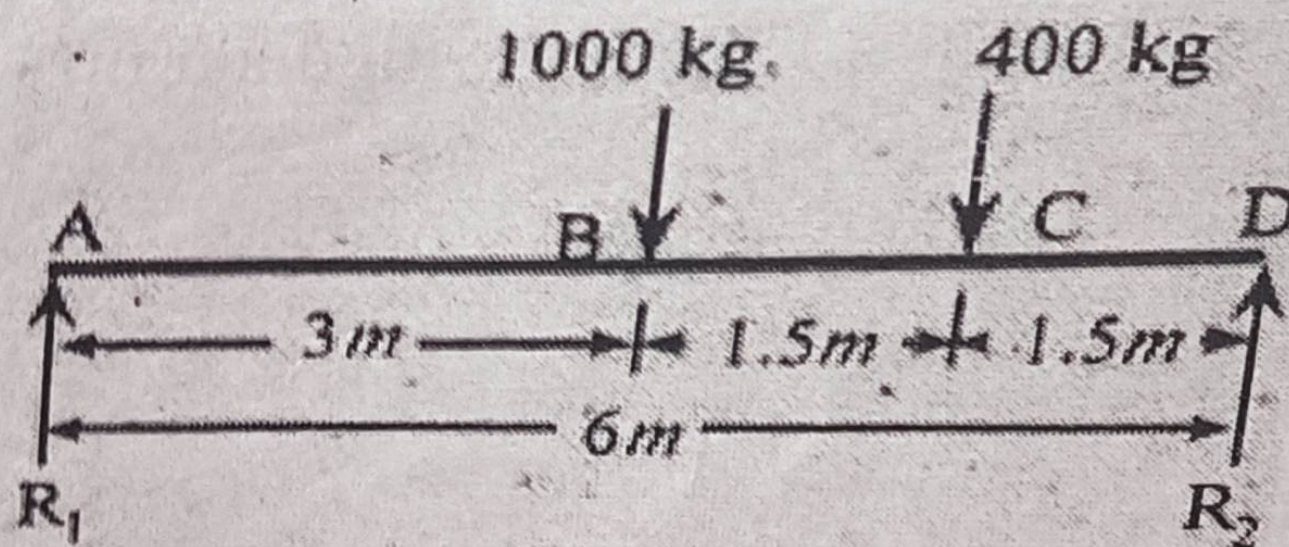
$$F_2^2 + 2F_1F_2\cos\theta$$

(b) The angle between the two forces of 80kN and 60kN is 60° , force being horizontal. Determine the magnitude and direction resultant force.



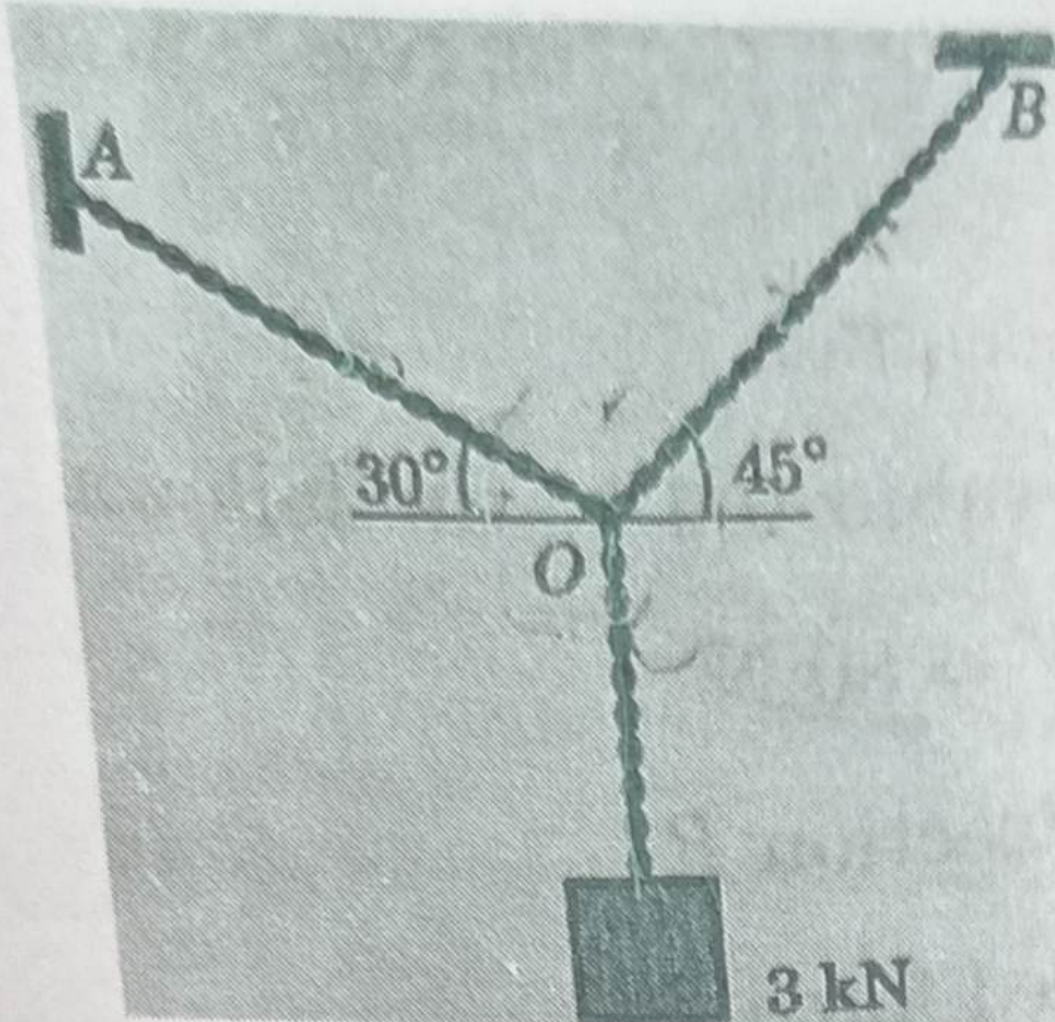
OR

Q.2 (a) Find the reaction forces R_1 and R_2 of given system.



(b) A machine weighing 3kN is supported by two ropes attached as shown in the figure. One of these ropes goes to the eye bolts in the wall and is inclined at 30° to the horizontal and other goes to the hook in ceiling and is inclined at 45° to the horizontal. Find the tensions in the two ropes.

(3)

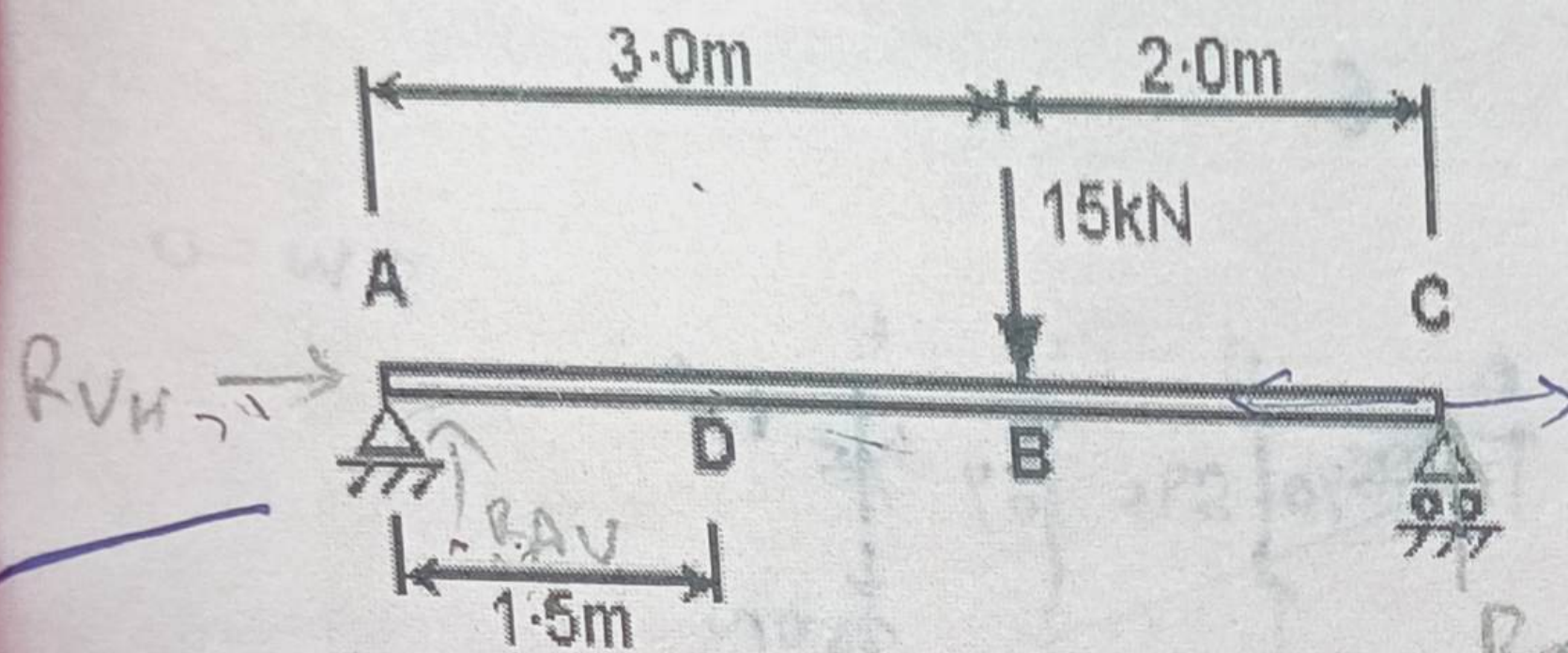


$$\frac{3}{\sin 30^\circ} = \frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 135^\circ}$$

$$\sum F_H = 0, \sum F_V = 0$$

$$\sum M = 0$$

- 1) Define the principle of virtual work and prove $\delta W = 0$. (3)



$$\sum F_H = 0$$

$$\rightarrow R_{FH} = 0$$

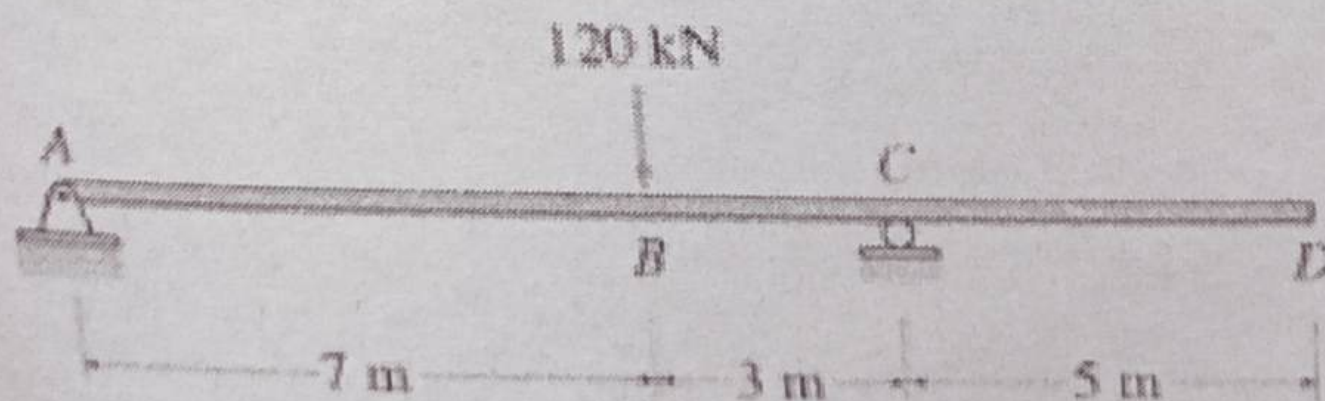
$$\sum F_V = 0$$

$$R_C = R_A$$

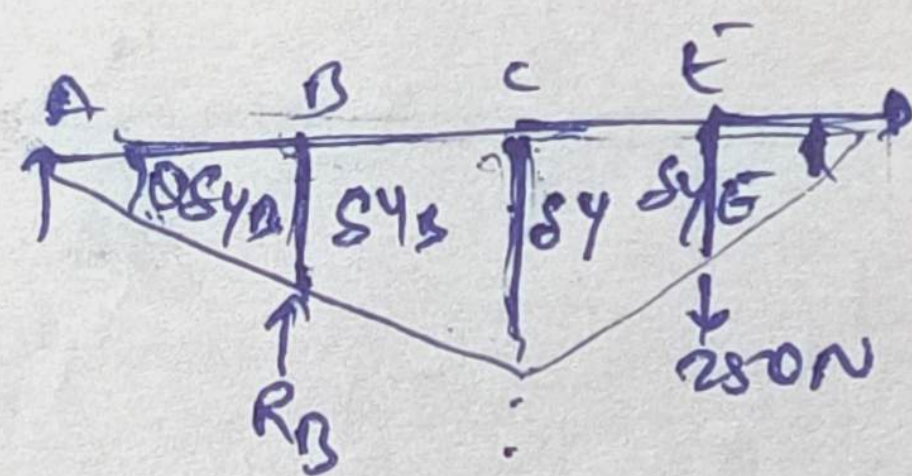
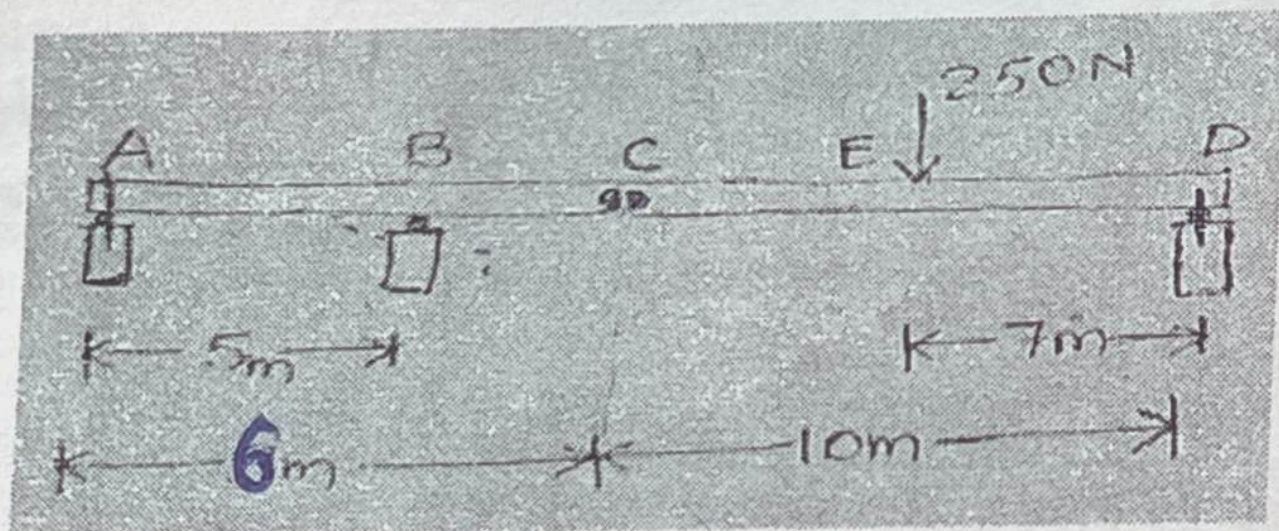
- (b) Find the reaction forces at point A and C of given system by principle of virtual work. (3)

OR

- Q.3 (a) Find the reaction forces at point A and C of given system by principle of virtual work. (3)



- (b) Two beams AC & CD of length 6m and 10m resp. are hinged at A and D. These are simply supported at B. Another support is given at point E. The loads on the beam are shown in fig. using the principle of virtual work find the reactions at point B. (3)



$$\sum W = 0$$

$$R_B \times \delta y_B - 250 \times \delta y_E = 0$$